EEPROM PROGRAM PROCEDURES

Step 1: Check the size of **EEPROM** flash

If the size of flash >64K, offset **308h** bit31 =1; else bit 31=0;

The below command is flash >64K , offset **308h is 80XX_XXXX**

If the size of flash <64K , offset **308h is 00XX_XXXX**

Step 2: EEPROM flash erase and check erase status

1. Erase command : Write offset **308h** ----80c7_0021
2. Check Erase status : Read offset **30Ch** ----check values of bit31~16. All zeros indicate erasing is successful.

   (Each Flash Vendor may be different for the read back data, please check Flash datasheet.)

Step 3: Write procedures

1. Write Enable : Write offset **308h**----8006_0021
2. Program Byte 0,1 :

   2-1 Write offset **30Ch** ---- XXXX(data)_XXXO(offset) O : 0,4,8,C

   2-2 Write offset **308h**----8002_0021

   2-3 Write Enable : Write offset **308h**----8006_0021
3. Program Byte 2,3 :

   3-1 Write offset **30Ch** ---- XXXX(data)_XXXO(offset) O: 2,6,A,E

   3-2 Write offset **308h**----8002_0021
4. Repeat from **Step1** to **Step3 to program another offset**

Step 4 : Read procedures (if needed)

1. Read Byte 0,1 :

   1-1 Write offset **30Ch** ---- XXXX (data ignore) _XXXO (offset) O: 0, 4, 8, C

   1-2 Write offset **308h**----8003_0021

   1-3 Read offset **30Ch** ---check bit 31 ~16(data)
2. Read Byte 2,3 :

   2-1 Write offset **30Ch** ---- XXXX (data ignore) _XXXO (offset) O: 2, 6, A, E

   2-2 Write offset **308h**----8003_0021

   2-3 Read offset **30Ch** ---check bit 31 ~16(data)

EEPROM CONTROL REGISTER – OFFSET 308h (Upstream Port Only)

BIT	FUNCTION	TYPE	DESCRIPTION	EEPROM/ I2C-SMBUS	DEFAULT
0	EEPROM Start	RW	Starts the EEPROM read or write cycle. 1b: start read or write cycle	Yes	0
3:1	Reserved	RsvdP	Not Support.	No	000b
4	EEPROM Autoload Status	RO	0b: EEPROM autoload was unsuccessfully or is disabled 1b: EEPROM autoload occurred successfully after PERST_L.	No	0
5	EEPROM is Audoload Disabled	RO	0b: EEPROM autoload is enabled 1b: EEPROM autoload is disabled	Yes	1
7:6	EEPROM Clock Rate	RW	Determines the frequency of the EEPROM clock which is derived from the primary clock. 00b: 500MHz/128 01b: 500MHz/(128*2) 10b: 500MHz/(128*4) 11b: 500MHz/(128*8)	Yes	00b
15:8	EEPROM Status	RO	Indicate the eeprom status.	No	00h
23:16	EEPROM Command	RW	01h: write STATUS register 02h: EEPROM write 03h: EEPROM read 04h: disable write operation 05h: read STATUS register 06h: enable write operation C7h: erase entire EEPROM	Yes	00h
30:24	Reserved	RsvdP	Not Support.	No	00h
31	Size 64K Mode	RW	0b: EEPROM size is less or equal to 64K 1b: EEPROM size is larger 64K	Yes	0

EEPROM ADDRESS AND DATA REGISTER – OFFSET 30Ch (Upstream Port Only)

BIT	FUNCTION	TYPE	DESCRIPTION	EEPROM/ I2C-SMBUS	DEFAULT
15:0	EEPROM Address	RW	Contains the EEPROM address.	Yes	0000h
31:16	EEPROM Data	RW	Contains the EEPROM data.	Yes	0000h